

A COMPARATIVE PHARMACOLOGICAL PROFILE OF DIBUCAINE HYDROCHLORIDE AND ANTIPSYCHOTICS

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ABSTRACT

Hematological parameters have an extreme degree of importance to perform physiological functions of the various systems in the body. Correct diagnosis of hematologic disorders is essential for rational therapy. Antipsychotics are prescribed for prolonged periods during which the risk of undesirable effects cannot be ruled out. The present study is designed to investigate the effects of antipsychotic drug like chlorpromazine, diazepam, clomipramine and spinal anesthetic dibucaine hydrochloride on few hematological parameters like clotting time, percentage of hemoglobin and erythrocyte sedimentation rate (ESR). Normal values of these parameters are also determined. The rabbits were kept for 12 days, the drugs like chlorpromazine (12.5 mg/1.5 kg of rabbit), diazepam (1.25 mg/1.5 kg of rabbit), dibucaine chloride (0.5 mg/1.5 kg of rabbit) and clomipramine (0.5 mg/1.5 kg of rabbit) were injecting daily. These studies indicated that different groups of antipsychotics have the same effects on hematological parameters, but the results of dibucaine hydrochloride are totally against of antipsychotics. This study would facilitate a better approach for the management of hematological problems in psychiatric patients and could be informative for the investigators for better, safe and selective use of antipsychotics in different psychiatric disorders.

INTRODUCTION

Several classes of drugs are effective in the treatment of psychosis. Antipsychotics are mostly used in the therapy of schizophrenia, organic psychosis, manic depressive illness and other acute idopathic psychotic illness. Their occasional use may be indicated in depression or in severe anxiety.

The use of antipsychotic agent is widespread, while the antipsychotic drugs have a revolutionary beneficial impact on medical and psychiatric practice, their effect on hematological parameters must be emphasized.

Dibucaine hydrochloride (cinchocaine) is used exclusively as spinal analgesic. It's a local anesthetic of amide type now generally used for surface anesthesia. It is one of the most potent and toxic of the long acting local

anesthetics and its parental use is restricted to spinal anesthesia. Epidural or spinal anesthesia may rarely result in paraplegia caused by an induced hematoma or abscess producing arachnoiditis, trauma or compression of the spinal cord. These blocks are therefore considered to be unsuitable for use in patients with pre-existing neurological disease, infection at the puncture site, or blood disorders or in those receiving low dose anticoagulant therapy to prevent postoperative deep vein thrombosis is still controversial (Stow and Burrows, 1987; Paranas and Schmidt, 1990).

Local anesthetics act by preventing the generation and transmission of impulses along nerve fibres and at nerve endings; depolarization and ion exchange are inhibited. The amide type local anesthetic can produce vasoconstriction (Aps and Reynolds, 1976).

In 1950 chlorpromazine was synthesized in France. The recognition of unique effects of chlorpromazine and its use in psychiatric patients marked the beginning of modern psychopharmacology. Swazey (1974) reported the use of chlorpromazine in psychomotor excitement and manic states. The effects of chlorpromazine is considerable degree of sedation, fall in blood pressure, tachycardia, decreased motor activity etc. Hence this pattern of response indicates a wide variety of central and peripheral actions of drug chlorpromazine and diazepam suppressed the rage reaction induced by intraperitoneal amphetamine sulphate in mice (Xu *et al.*, 1992). One of the most disturbing side effect of chlorpromazine "AKATHISIA" is manifested by a marked change in motor activity. It inhibits the reuptake of norepinephrine and 5-HT in rat cerebral cortex. Chlorpromazine inhibited 5-HT uptake into platelet from normal control in a competitive manner (Arora and Meltzer, 1983). Chlorpromazine and other antipsychotics binds significantly to membranes and to plasma proteins over 85% of drug (Sunderland and Cohen, 1987).

In present study the effects of dibucaine (cinchocaine) compared with antipsychotics like chlorpromazine, clomipramine and diazepam for their pharmacological profile based on hematological parameters. Antipsychotics used for prolong period of time due to which our object is to determine the effects of these drugs.

MATERIALS AND METHOD

In present study rabbits weighing 0.85-1.50 kg were used. These were divided into 5 batches first, then each batch of 5 (n=5) rabbits for each drug and one batch of 5 (n=5) rabbits was used as control group.

Animals were obtained from Aga Khan University. All procedures were carried out

with the care and use of laboratory animals and with animal study protocol. Rabbits were given food, water ad libitum.

Hematological parameters were observed by chronic administration of drugs on male rabbits. We kept the rabbits for 12 days, animals were injected intramuscularly according to the following dosage manner.

- (i) Dibucaine hydrochloride (DBN) (cinchocaine) 0.5 mg/1.5 kg of rabbits.
- (ii) Chlorpromazine (CPZ) given 12.5 mg/1.5 kg of rabbits.
- (iii) Clomipramine (CMP) given 0.5 mg/1.5 kg of rabbits.
- (iv) Diazepam (BDZ) given 1.25 mg/1.5 kg of rabbits.

They were sacrificed after 12 days in the manner of chronic administration of drug. Then the effects of antipsychotics were observed on the following hematological parameters.

ESR:

ESR was determined by Westergren method 0.2 ml 3.8% sodium citrate solution was taken in a graduated tube. Then the blood was added to raise the volume to 1 ml. The mixture was shaken gently and blood was sucked into a pipette 2.5 mm in diameter and graduated to 200 mm. This pipette clamped in a stand and sealed by being clipped on to a cork at the bottom finally the distance settled in millimeters in one hour was noted.

Hemoglobin Percentage:

In present study for estimation of hemoglobin concentration we used hemoglobin meter with Sahli's pipette.

Immediately transfer the blood into a test tube containing 5 ml of hemoglobin reagent. After diluting the blood with 5 ml hemoglobin reagent, adjust the hemoglobin meter to zero by distilled water. At last pour all the diluted

blood into drain, and note the concentration of hemoglobin in percentage and also in gram/dl.

Coagulation or Clotting Time:

Clotting time or coagulation time was observed by taking the blood in a petri dish. The blood was lifted with the help of a capillary tube until the fibre formation started and the time was noted in minutes or seconds.

RESULTS

In the present study antipsychotic chlorpromazine, antidepressant clomipramine and anxiolytics like diazepam had some effects on clotting time, percentage of hemoglobin and on erythrocyte sedimentation rate. Normal values of above mentioned hematological parameters have also been determined significance (probability) is calculated by student's "t" test.

In present study almost all antipsychotics have decreased, the ESR. The ESR in control is found as 1.92 mm/hr.

CPZ:

Has decreased the ESR which is statistically significant ($P < 0.01$).

BDZ:

Statistically significant ($P < 0.01$) reduction in erythrocyte sedimentation rate.

CMP:

There is statistically significant ($P < 0.01$) reduction in ESR after CMP administration.

DBN:

Table 1 and Fig. 1 shows statistically significant ($P < 0.01$) increased in ESR after giving DBN.

Table 1 and Fig. 1 shows the change in erythrocyte sedimentation rate in test animals ($n=5$) when compared with control animals ($n=5$).

All antipsychotics have shown to increase the clotting time as compared to control. The controlled animals have taken 4.70 minutes for their blood to clot.

CPZ:

Significant increase in coagulation time is observed which is statistically significant ($P < 0.01$).

BDZ:

Diazepam has increased the clotting time significantly ($P < 0.01$).

CMP:

Statistically significant ($P < 0.1$) increase.

DBN:

Dibucaine has produced a significant decrease ($P < 0.01$) in clotting time.

Table 1 and Fig. 2 shows the clotting time calculated from controlled animals ($n=5$) and test animals after the administration of the drugs.

In present study, the percentage of hemoglobin is increased by all antipsychotics and reduced by local anesthetic. The percentage of hemoglobin in control animal is found as 80.60%.

CPZ:

Statistically significant ($P < 0.01$) increase the percentage of hemoglobin is noted.

BDZ:

Significant increase ($P < 0.01$) in the percentage of hemoglobin after the administration of diazepam.

CMP:

Clomipramine has increased the percentage of hemoglobin significantly ($P < 0.01$).

DBN:

Table 1 shows statistically significant ($P < 0.01$) decrease in percentage of hemoglobin after giving dibucaine hydrochloride.

Table 1 and Fig. 3 shows the change in percentage of hemoglobin in test animals ($n=5$) when compared with control animals ($n=5$).

DISCUSSION

In the light of the present study results, it clearly shows that DBN decreased the percentage of hemoglobin significantly and as a resultant effect ESR was found elevated significantly. A number of possible conclusions are suggested. Dibucaine which is a local anesthetic has effect on membrane fluidity, this phenomenon is studied by Gjonnaess *et al.*, (1992). They reported that platelets when incubated with dibucaine then fluidity was increased with increased platelet aggregation. It may be suggested that increased fluidity could lead to rupture of red blood cells (RBCs) membrane. Thus hemolysis can be increased leading to low hemoglobin percentage and elevation of ESR. The effect on ESR may be through activation or suppression of phosphokinases. Dibucaine has effect on activation of kinases (Goodman and Gillman, 1996). There is another supporting evidence reported by Atzori *et al.* (1992) that the microsomal Ca^{2+} ATPase inhibitor 2,5-di-(tert-butyl)-benzohydroquinone (tBuBHQ) induced bronchioconstriction and vasoconstriction. These effects were accompanied by increased levels of thromboxane and prostacycline. The involvement of a phospholipase in the generation of thromboxane A_2 (TxA_2) was supported further by dibucaine's protection on tBuBHQ effect. Atzori *et al.* Indicated that tBuBHQ, acts probably by inhibiting the microsomal Ca^{2+} ATPase, inhibition of ATPase can trigger the arachidonic acid cascade leading to the formation of ATPase and vasoconstriction. In present study dibucaine significantly decreased the clotting

time due to the vasoconstrictor property of amide anesthetic, which is also reported by Willalts and Reynold (1985). Another supporting evidence for our results reported by Hultin *et al.* (1992) that dibucaine treated platelets increased the peak of factor VIII clotting activity, may lead to significantly decreased clotting time. Wolosker *et al.* (1992) reported that local anesthetic like dibucaine increased the Ca^{2+} efflux through the nonenergized state of the Ca^{2+} ATPase, rather than promoting a non-specific Ca^{2+} leakage through the membrane. This effect of dibucaine could be related to effect on hemoglobin, ESR and duration of clotting.

In present study results of hematological parameters suggests that antipsychotic chlorpromazine, antidepressant clomipramine and anxiolytics like diazepam had same effect on clotting time, percentage of hemoglobin and on erythrocyte sedimentation rate.

Ohsuka *et al.* (1995) reported that both clomipramine and chlorpromazine had inhibitory effects on 3H-ketanserin binding, 5-HT induced intracellular Ca^{2+} increase and 5-HT induced shape change in human platelets and causes the effect on the 5-HT₂ receptor mediated signal transducing system in human platelets. Its postulated that present results of hematological parameters due to the inhibitory effects of CMP and CPZ on 5-HT induced intracellular Ca^{2+} increase, and to which structural and functional integrity of platelets, are maintained may be due to 5-HT induced change in platelets and vessel wall hemostasis is also maintained and could lead to an increase in clotting time. Increase in percentage of hemoglobin by these psychotropic drugs may be due to the effect of membrane fluidity. Tanji *et al.* (1992) reported that chlorpromazine was shown to decrease the membrane fluidity. Decreased fluidity could lead to the prevention of RBCs membrane hemolysis and thus leading to an increase in hemoglobin percentage and automatically decreased the ESR and these

effects are mediated possibly through effect on membrane fluidity.

Superoxide formation inhibition by CPZ is also helpful in erythropoiesis (Fischer and Cross, 1977; Ikeda, 1981) may leads to prevent hemolysis and increase percentage of hemoglobin and decrease ESR. Superoxide inhibition is also a factor for the mechanism of erythropoiesis. As reported by Beitner *et al.* (1989) chlorpromazine and other psychotropic drugs increase the level of ATP

releaser. There is a link between cocaine analogs and antidepressants that antidepressant therapy has been attempted in the treatment of cocaine abusers, and the data collected with these drugs have suggested that they may be useful therapeutic tool (Giannini and Billett, 1987; Gawin *et al.*, 1989).

Psychotropic drugs are prescribed for prolong use during which several unwanted effects are produced, therefore a great deal of attention is needed. Drugs may adversely

Table – 1

Table shows ESR (erythrocyte sedimentation rate), clotting time and hemoglobin percentage.

		ESR	Clotting time	Hemoglobin percentage
1.	Control	1.92±0.09	4.70±0.25	80.60±0.80
2.	Dibucaine hydrochloride	2.96±0.08**	1.02±0.04**	52.00±2.40**
3.	Benzodiazepine	1.08±0.09**	4.98±0.04**	112.60±0.48**
4.	Chlorpromazine	1.40±0.12**	4.96±0.08**	114.20±0.97**
5.	Clomipramine	1.08±0.09**	4.98±0.04**	110.60±0.80**

Values represents means ± S.D. (n=5). Difference with respect to control was evaluated by student "t" test (**P<0.01).

and induces enzymes such as hexokinase and glucose-6-phosphate dehydrogenase. The enhancement in metabolism results in exaggerated formation of reduced equivalents (NADPH₂) which maintain an erythropoietin tripeptide i.e. glutathione. Maintenance of glutathione is a clear indication of increased erythropoiesis, and maintain shape and hemolysis of RBCs. Prevention of hemolysis by these psychotropic drugs may play a role in reduction of ESR and increment in hemoglobin percentage.

This work may further be extended to see the interactions of different classes of antipsychotics, side effects caused by different amines uptakes inhibitors or amine

affect various physiological systems. Selection of drug depends upon genetic, biochemical, hematological, neurochemical, behavioural and environmental factors of patients. These drugs should be used with great caution in patients with cardiovascular disease, inflammatory and hepatic disease, different previous pathological problems because these drugs affects mainly the neurochemistry of brain and also affects hematological and biochemical parameters of patients.

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